Roll No.:

Test Date: 09-03-2025

Aakash
Medical | IIT-JEE | Foundations

E
Booklet Code

AR-008B

AIATS
ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2025

National Eligibility-cum-Entrance Test (NEET)

TEST No. 6

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 6

TOPICS OF THE TEST

Physics	Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current, Electromagnetic Waves
Chemistry	Amines, The *d* & *f*-Block Elements, Coordination Compounds
Botany	Molecular basis of inheritance, Microbes in human welfare
Zoology	Evolution, Human Health and Disease

MM : 720

TEST - 6

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

1. The ratio of radii of two loops is 1 : 2 and the ratio of current in them is 2 : 1. Then the ratio of their respective magnetic moment is
   (1) 1 : 2
   (2) 2 : 1
   (3) 1 : 4
   (4) 4 : 1

2. A uniform electric and magnetic field are applied along the direction of velocity of a proton. The proton will
   (1) Continue to move in straight line with uniform speed
   (2) Continue to move in straight line with uniform acceleration
   (3) Move in uniform circular motion
   (4) Move in non-uniform circular motion

3. Choose the correct expression for Biot-Savart's law for a current carrying element (where symbols have their usual meanings).
   (1) $\frac{\mu_0}{2\pi}\frac{i\left(d\vec{l}\times\vec{r}\right)}{r^2}$
   (2) $\frac{\mu_0}{4\pi}\frac{i\left(d\vec{l}\times\vec{r}\right)}{r^2}$
   (3) $\frac{\mu_0}{2\pi}\frac{i\left(d\vec{l}\times\vec{r}\right)}{r^3}$
   (4) $\frac{\mu_0}{4\pi}\frac{i\left(d\vec{l}\times\vec{r}\right)}{r^3}$

4. Magnetic moment of a bar magnet is M. If it is cut into three equal parts perpendicular to its length, then
   (1) Its pole strength and magnetic moment both becomes $\frac{1}{3}$ times
   (2) Its pole strength remains same but the magnetic moment becomes $\frac{1}{3}$ times
   (3) Its pole strength becomes $\frac{1}{3}$ times but magnetic moment remains same
   (4) Its pole strength becomes 3 times and magnetic moment becomes $\frac{1}{3}$ times

5. If B_1 is the magnitude of magnetic field at a distance r on the axis of a short bar magnet and B_2 is the magnitude of magnetic field at a distance r on the equatorial line of that bar magnet, then
   (1) $B_1 = B_2$
   (2) $B_1 = 2B_2$
   (3) $B_1 = \frac{B_2}{2}$
   (4) $B_2 = 0$

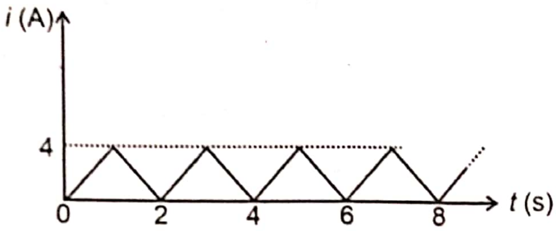
6. Lenz's law states that the direction of induced current in a conducting coil is such that it
   (1) Opposes the increase of magnetic flux through the coil only
   (2) Opposes the decrease of magnetic flux through the coil only
   (3) Opposes the change in magnetic flux through the coil
   (4) Supports the change in magnetic flux through the coil

Space for Rough Work

7. The magnetic flux (ϕ_B) through a rectangular loop varies with time (t) as $\phi_B = (t^2 + 2t - 2)$ Wb where t is in seconds. The magnitude of average emf induced in the loop between $t = 1$ s to $t = 4$ s is

(1) 3 V
(2) 5 V
(3) 7 V
(4) 9 V

8. A time varying current is shown in the diagram given below. The average value of current is



(1) 2 A
(2) 4 A
(3) 2.4 A
(4) 3.6 A

9. In a series R–L AC circuit, the phase difference between the source voltage and current is

(1) $\frac{\pi}{2}$
(2) π
(3) Greater than zero and less than $\frac{\pi}{2}$
(4) Greater than $\frac{\pi}{2}$ and less than π

10. A 100 μF capacitor is connected to an AC supply having angular frequency ω = 100 rad s^{-1}. The reactance of the capacitor is

(1) 10 Ω
(2) 1 Ω
(3) 0.1 Ω
(4) 100 Ω

11. When a series RLC circuit is at resonance then

(1) Current in the circuit is maximum but impedance of the circuit is minimum
(2) Current in the circuit is minimum but impedance of the circuit is maximum
(3) Current in the circuit is maximum and net impedance of the circuit is zero
(4) Both (1) and (3)

12. A DC source of 220 V is applied to primary windings of an ideal transformer having transformer ratio 4:1, the output voltage of the transformer from the secondary is

(1) 55 V
(2) 880 V
(3) Zero
(4) 220 V

13. An EM wave is propagating through free space where B_0 and E_0 are the amplitude of magnetic field and electric field respectively. The average energy density associated with magnetic field is

(1) $\frac{1}{2\mu_0}B_0^2$
(2) $\frac{1}{4\mu_0}B_0^2$
(3) $\frac{1}{2E_0}B_0^2$
(4) $\frac{B_0^2}{4E_0}$

14. The phase difference between oscillating electric and magnetic field for an EM wave is

(1) Zero
(2) $\frac{\pi}{2}$
(3) π
(4) $\frac{\pi}{4}$

15. For an EM wave travelling along x-axis, the electric field is oscillating along y-axis with amplitude $E_0 = 45\times10^{-2}$ V/m. The magnetic field is oscillating along

(1) z-axis with amplitude 1.5 nT
(2) y-axis with amplitude 1.5 nT
(3) z-axis with amplitude 10.5 nT
(4) y-axis with amplitude 10.5 nT

Space for Rough Work

16. Given below are two statements:

Statement I : A time varying magnetic field can produce a time varying electric field.

Statement II : An electric charge oscillating harmonically with frequency *f*, produces electromagnetic waves of the same frequency *f*.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but statement II is correct

17. In the given columns the electromagnetic waves and some of their applications are mentioned in column I and column II respectively. Match the column I and II and choose the correct option.

	Column I		Column II
(A)	Microwaves	(P)	Remote switches of household electronics systems.
(B)	Radio waves	(Q)	To kill germs in water purifiers
(C)	Infrared waves	(R)	Radar system for aircraft navigation
(D)	Ultraviolet rays	(S)	Radio, FM broadcasting

(1) (A)→R, (B)→S, (C)→Q, (D)→P

(2) (A)→S, (B)→R, (C)→Q, (D)→P

(3) (A)→R, (B)→S, (C)→P, (D)→Q

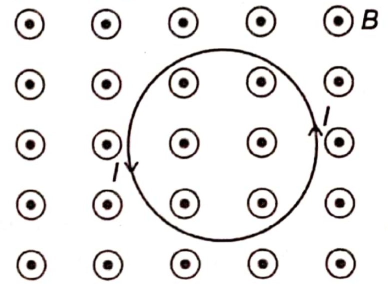
(4) (A)→S, (B)→R, (C)→P, (D)→Q

18. The magnetic flux increases from 4 Wb to 12 Wb through a coil of resistance 200 Ω in 0.2 second. The net charge passed through the coil is

(1) 20 mC (2) 40 mC

(3) 10 mC (4) 80 mC

19. A flexible metallic ring carrying current *I* in anticlockwise direction placed in transverse magnetic field (*B*) is as shown. The radius of the ring



(1) Will increase

(2) Will decrease

(3) Will remain unchanged

(4) May increase or decrease

20. The magnetic susceptibility of diamagnetic material is

(1) Equal to zero (2) Less than zero

(3) Greater than zero (4) Equal to one

21. In a series LCR circuit the resonant frequency is 500 Hz and half power points have frequencies as 475 Hz and 525 Hz. The quality factor of the circuit is

(1) 10 (2) 5

(3) 0.1 (4) 0.2

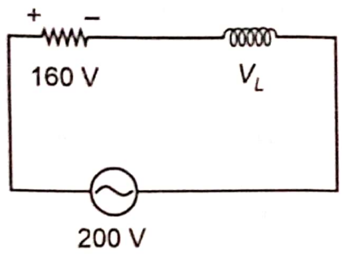
22. An AC voltage is represented by $E = 220\sqrt{2}\sin(100\pi t)\,\text{V}$. The number of times current will become zero in 2 seconds

(1) 50 times (2) 100 times

(3) 200 times (4) 400 times

Space for Rough Work

23. In the circuit shown below, the rms value of voltage across inductor is



(1) 100 V

(2) 120 V

(3) 160 V

(4) 80 V

24. A solenoid has a core of a material with relative permeability 200. The windings of the solenoid is insulated from the core and carries a current of 1 A. The net magnetic field intensity (B) at the centre of solenoid if the number of turns per unit length is 2000 m^{-1}

(1) 0.16π T

(2) 1.6π T

(3) 0.8π T

(4) 0.8π mT

25. For permanent magnets, the material has

(1) High retentivity and low coercivity

(2) High coercivity and low retentivity

(3) High retentivity and high coercivity

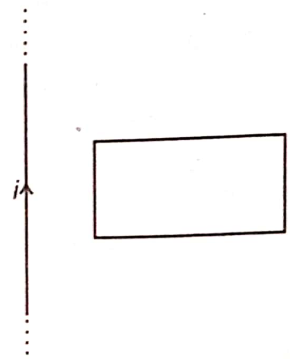
(4) Low retentivity and low coercivity

26. At resonance, the value of power factor in an LCR series circuit is

(1) Zero (2) 1

(3) $\frac{1}{2}$ (4) $\frac{1}{4}$

27. A rectangular conducting loop is placed in the neighbourhood of a coplanar long straight wire carrying a current i. If current is decreasing, then the induced current in the loop is



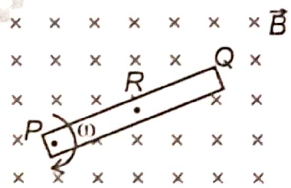
(1) In clockwise direction

(2) In anticlockwise direction

(3) Zero

(4) First anticlockwise and then clockwise

28. A uniform conducting rod of length L is rotated about one of its end P with angular velocity ω in a transverse magnetic field $\vec{B}$ as shown. The magnitude of potential difference between the mid-point (R) and free end (Q) is

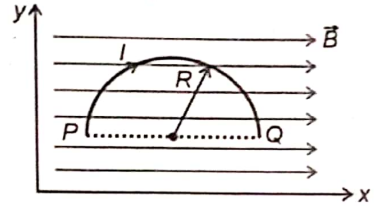


(1) $\frac{1}{2}B\omega L^2$ (2) $\frac{1}{8}B\omega L^2$

(3) $\frac{1}{4}B\omega L^2$ (4) $\frac{3}{8}B\omega L^2$

Space for Rough Work

29. A conducting semicircular loop of radius R having current I is placed in xy plane. There exists a uniform magnetic field $\vec{B}$ along x-axis in the space as shown. If the loop is released from rest then its acceleration



(1) Is along x-axis (2) Is along y-axis
(3) Is variable (4) Is zero

30. Electric field (E_y) and magnetic field (B_z) associated with an EM wave are given by following equation

$\vec{E}_y = -E_0 \sin(\omega t + kx)\hat{j}$

$\vec{B}_z = B_0 \sin(\omega t + kx)\hat{k}$

Then,

(1) The wave is travelling along $+x$ direction and $\frac{\omega}{k} = \frac{E_0}{B_0}$

(2) The wave is travelling along $-x$ direction and $\frac{\omega}{k} = \frac{E_0}{B_0}$

(3) The wave is travelling along $-x$ direction and $\frac{\omega}{k} = \frac{B_0}{E_0}$

(4) The wave is travelling along $+x$ direction and $\frac{\omega}{k} = \frac{B_0}{E_0}$

31. Electromagnetic wave of intensity I falls normally on a perfectly reflecting surface of area A. If c is the speed of the EM wave then average force on the surface is

(1) $\frac{I}{c}$ (2) $\frac{2I}{c}$
(3) $\frac{IA}{c}$ (4) $\frac{2IA}{c}$

32. The magnetic energy stored in an inductor of inductance 2 μH carrying a current of 4 A is

(1) 4 μJ (2) 8 μJ
(3) 12 μJ (4) 16 μJ

33. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): For long distance transmission of electric energy, generally AC is used.

Reason (R): It is easier for AC to step up and transmitted as high voltage (low current) signals.

In the light of the above statements, the correct option is

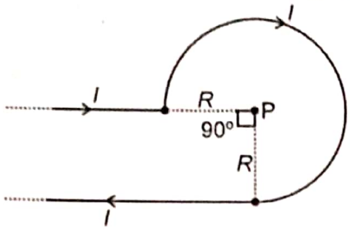
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

34. A galvanometer having resistance 50 Ω and full deflection current 10 mA is converted into voltmeter of range 0–5 V. Then

(1) 450 Ω resistance is connected in series with galvanometer
(2) 450 Ω resistance is connected in parallel with galvanometer
(3) 4950 Ω resistance is connected in series with galvanometer
(4) 4950 Ω resistance is connected in parallel with galvanometer

Space for Rough Work

35. An infinite wire has a circular bend of radius R and carrying current I as shown in figure. The magnitude of magnetic field at the centre P is given by



(1) $\frac{\mu_0 I}{2R}\left[\frac{3}{2}+\frac{1}{\pi}\right]$ (2) $\frac{\mu_0 I}{4R}\left[\frac{3}{2}+\frac{1}{\pi}\right]$

(3) $\frac{\mu_0 I}{4R}\left[\frac{-3}{2}+\frac{1}{\pi}\right]$ (4) $\frac{\mu_0 I}{2R}\left[\frac{3}{2}-\frac{1}{\pi}\right]$

36. A particle having charge q and mass m moves in a circular orbit of radius r with angular speed ω. The ratio of magnitude of its magnetic moment to that of angular momentum is

(1) $\frac{q}{2m}$ (2) $\frac{q}{m}$

(3) $\frac{2q}{m}$ (4) $\frac{2q}{mr}$

37. A charged particle q having mass m is projected with velocity $(\vec{v})$ in a uniform magnetic field $(\vec{B})$ such that angle between $\vec{v}$ and $\vec{B}$ is θ, where $0° < \theta < 90°$. Then path of the charge particle is

(1) Straight line

(2) Circle

(3) Helical path with constant pitch

(4) Helical path of variable pitch

38. If current sensitivity of a moving coil galvanometer is 10 degree/mA and voltage sensitivity is 5 degree/V, then resistance of the galvanometer is

(1) 500 Ω (2) 1000 Ω

(3) 2000 Ω (4) 4000 Ω

39. A charged particle q having mass m is projected with velocity $\vec{v}$ in a uniform magnetic field $(\vec{B})$ such that the velocity $(\vec{v})$ is perpendicular to magnetic field $(\vec{B})$. Then

(1) Force on the charged particle is constant

(2) Momentum of charged particle is constant

(3) Kinetic energy of the charged particle remains constant

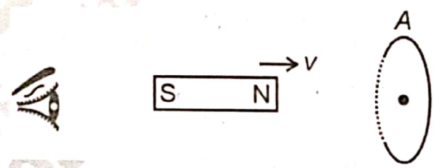
(4) Work done by magnetic force on charged particle is non-zero

40. A charged particle of mass 10^{-3} kg and charge 10^{-5} C enters into a space where electric field E of strength 50 V/m and magnetic field B of 10^{-4} T exist perpendicular to each other. If charge crosses the space without any deviation then velocity of the charge is

(1) 5×10^{8} m s^{-1} (2) 5×10^{5} m s^{-1}

(3) 5×10^{6} m s^{-1} (4) 5×10^{4} m s^{-1}

41. A bar magnet moves towards a conducting loop with constant velocity v as shown.



As seen from the side of magnet, the direction of

(1) Induced current in the loop is clockwise

(2) Induced current in the loop is anticlockwise

(3) No current is induced in the loop

(4) At first the induced current is clockwise and then it would become anticlockwise

42. The dimensional formula for mutual inductance is

(1) $[ML^2T^{-2}A^{-1}]$ (2) $[ML^2T^{-2}A^{-2}]$

(3) $[MLT^{-2}A^{-2}]$ (4) $[ML^2T^2A^{-2}]$

Space for Rough Work

43. In a series LCR circuit, the phase difference between voltage across inductor and voltage across capacitor is

(1) 0°
(2) 45°
(3) 90°
(4) 180°

44. Electromagnetic waves are produced by

(1) A static charge
(2) Charge moving with constant velocity
(3) An accelerating charge
(4) Both (1) and (2)

45. A capacitor of capacitance C is connected across an AC source of voltage V, given by $V = V_0\cos\omega t$.

Then displacement current between the plates of the capacitor would be given by

(1) $\omega CV_0\sin\omega t$
(2) $-\omega CV_0\sin\omega t$
(3) $\omega CV_0\cos\omega t$
(4) $-\omega CV_0\cos\omega t$

[CHEMISTRY]

46. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A) : Aromatic primary amines can be prepared by the Gabriel phthalimide synthesis.

Reason (R) : In Gabriel phthalimide synthesis, aryl halide undergo nucleophilic substitution with the anion formed by phthalimide.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

47. Arrange the following compounds in their decreasing order of boiling point

$n-C_4H_9NH_2$ (a) $C_2H_5N(CH_3)_2$ (b) $(C_2H_5)_2NH$ (c)

(1) (a) > (b) > (c)
(2) (b) > (c) > (a)
(3) (a) > (c) > (b)
(4) (b) > (a) > (c)

48. Total number of bridging CO groups in $[Mn_2(CO)_{10}]$ and $[Co_2(CO)_8]$ respectively are

(1) Zero and two
(2) Four and zero
(3) Two and zero
(4) Zero and four

49. Given below are two statements.

Statement I: In $[Ni(CO)_4]$, the Ni – C σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of the Nickel.

Statement II: In $Ni(CO)_4$, the Ni – C π bond is formed by the donation of a pair of electrons from a filled *d* orbital of nickel into the vacant antibonding π* orbital of carbon monoxide.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

Space for Rough Work

50. The structure of $[Fe(CO)_5]$ and oxidation state of central atom in $[Fe(CO)_5]$ respectively are

(1) Trigonal bipyramidal and +5

(2) Pentagonal bipyramidal and +5

(3) Square pyramidal and zero

(4) Trigonal bipyramidal and zero

51. Given below are two statements.

Statement I : $[Co(NH_3)_6]^{3+}$ is an inner orbital complex whereas $[Co(H_2O)_6]^{3+}$ is an outer orbital complex.

Statement II : $[Co(NH_3)_6]^{3+}$ is diamagnetic whereas $[Co(H_2O)_6]^{3+}$ is paramagnetic

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

52. Match List-I with List-II

	List-I (Complex)		**List-II (Hybridisation)**
a.	$[NiCl_4]^{2-}$	(i)	d^2sp^3
b.	$[Co(C_2O_4)_3]^{3-}$	(ii)	sp^3
c.	$[Ni(CN)_4]^{2-}$	(iii)	dsp^2
d.	$[CoF_6]^{3-}$	(iv)	sp^3d^2

Choose the correct option

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(i), c(iii), d(iv)

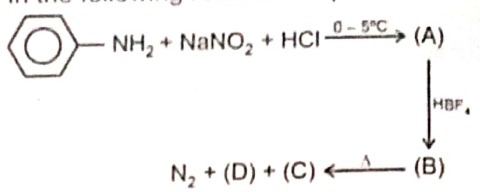
(3) a(iii), b(iv), c(ii), d(i) (4) a(iii), b(i), c(ii), d(iv)

53. Which of the following species absorbs light of minimum wavelength?

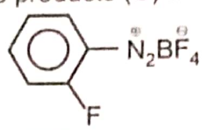
(1) $[Co(NH_3)_6]^{3+}$ (2) $[Co(CN)_6]^{3-}$

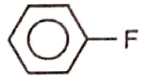
(3) $[CoCl(NH_3)_5]^{2+}$ (4) $[Co(NH_3)_5H_2O]^{3+}$

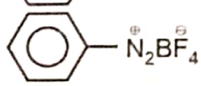
54. In the following reaction sequences

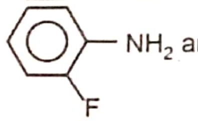


The products (C) and (D) are

(1)  and F_2

(2)  and BF_3

(3)  and F_2

(4)  and BF_3

55. The correct order of increasing ligand field strength is

(1) $S^{2-} < OH^- < en < edta^{4-}$

(2) $OH^- < S^{2-} < edta^{4-} < en$

(3) $S^{2-} < OH^- < edta^{4-} < en$

(4) $en < S^{2-} < OH^- < edta^{4-}$

56. Given below are two statements.

Statement I : Ligands for which $\Delta_0 < P$ are known as strong field ligands

Statement II : Ligands for which $\Delta_0 > P$ are known as weak field ligands

In the light of above statements choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

Space for Rough Work

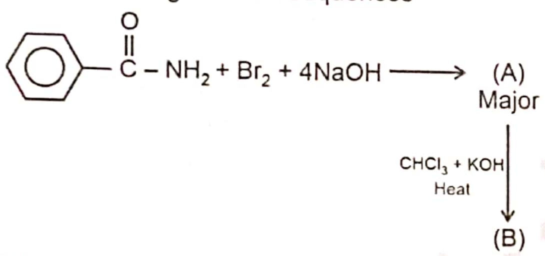
57. Match List-I with List-II

	List-I Species		List-II Number of unpaired electron(s) in central atom
a.	$[MnCl_6]^{3-}$	(i)	0
b.	$[Ni(CO)_4]$	(ii)	1
c.	$[Fe(CN)_6]^{3-}$	(iii)	2
d.	$[Mn(CN)_6]^{3-}$	(iv)	4

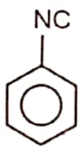
Choose the correct option

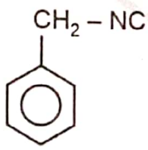
(1) a(iv), b(ii), c(i), d(iii)
(2) a(iv), b(i), c(ii), d(iii)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(iv), b(iii), c(ii), d(i)

58. In the following reaction sequences

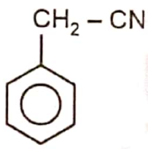


The major product (B) will be

(1) 

(2) 

(3) 

(4) 

59. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A): Geometrical isomerism is not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion.

Reason (R): In tetrahedral complexes, the relative positions of the unidentate ligands attached to the central metal atom are the same with respect to each other.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

60. Choose the incorrect match

	Compounds		IUPAC name
(1)	$[Co(NH_3)_4(H_2O)Cl]Cl_2$	–	Tetraammineaquachloridocobalt (III) chloride
(2)	$[Co(NH_3)_5(CO_3)]Cl$	–	Pentaamminecarbonatocobalt (III) chloride
(3)	$[PtCl_2(en)_2](NO_3)_2$	–	Bis (ethane-1, 2-diamine) dichloridoplatinum (IV) nitrate
(4)	$K_3[Fe(C_2O_4)_3]$	–	Potassium trioxalatoferrate (III)

61. Given below are two statements.

Statement I : The UV-visible absorption bands in the spectra of lanthanoid ion are broad, probably because of the excitation of electrons involving f orbitals.

Statement II : Zr and Hf have almost identical radii due to lanthanoid contraction.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

Space for Rough Work

62. Consider the following statements

(a) Lanthanoids liberate hydrogen gas on reaction with dilute acids

(b) Ionic radii of Ho^{3+} is lesser than Tm^{3+}

(c) Bronze is an alloy of copper and zinc

(d) In aqueous solution, Cu(I) compounds are unstable and undergo disproportionation

The correct statements(s) is/are

(1) (a) only (2) (b) and (d) only

(3) (b) and (c) only (4) (a) and (d) only

63. Calculate the CFSE for $[CoF_6]^{3-}$ ion. (ignoring pairing energy)

(1) $-0.4\ \Delta_0$ (2) $-1.6\ \Delta_0$

(3) $-0.6\ \Delta_0$ (4) $-1.2\ \Delta_0$

64. One mole of which octahedral complex of cobalt will give 3 moles of AgCl as precipitate when reacts with excess of $AgNO_3$ solution?

(1) $CoCl_3{\cdot}6NH_3$ (2) $CoCl_3{\cdot}4NH_3$

(3) $CoCl_3{\cdot}3NH_3$ (4) $CoCl_3{\cdot}5NH_3$

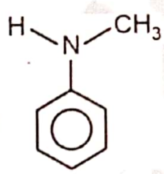
65. The highest negative standard reduction potential $\left(E^{\circ}_{M^{2+}/M}\right)$ among the following is of

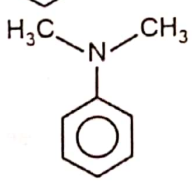
(1) Ti (2) Cr

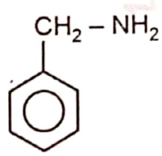
(3) Mn (4) Co

66. Which of the following is most basic in aqueous phase?

(1) 

(2) 

(3) 

(4) 

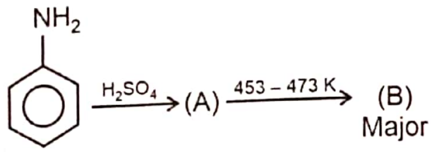
67. Correct order of C – O bond strength of the given species is

$[V(CO)_6]^-$ (i) $[Mn(CO)_6]^+$ (ii) $[Cr(CO)_6]$ (iii)

(1) (ii) > (iii) > (i) (2) (i) > (iii) > (ii)

(3) (ii) > (i) > (iii) (4) (iii) > (i) > (ii)

68. In the following reaction



The product (B) will be

(1) Sulphanilic acid

(2) Anilinium hydrogensulphate

(3) Benzenesulphonic acid

(4) o-aminobenzenesulphonic acid

69. Which of the following transition metal ions are coloured in aqueous medium?

(a) Ti^{4+} (b) Cr^{3+}

(c) Co^{3+} (d) Mn^{2+}

The correct option is

(1) (a) and (b) only (2) (b) and (c) only

(3) (b), (c) and (d) only (4) (a) and (d) only

70. Consider the following statements about interstitial compounds

(a) They are usually non-stoichiometric

(b) They retain metallic conductivity.

(c) They have lower melting points, lower than those of pure metals.

(d) They are chemically inert

The correct statements are

(1) (a) and (d) only

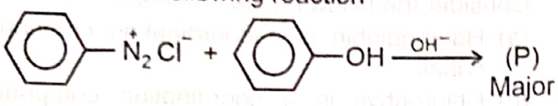
(2) (a), (b) and (c) only

(3) (b) and (c) only

(4) (a), (b) and (d) only

Space for Rough Work

71. Consider the following reaction



The colour and name of the product (P) will be

(1) Yellow and o-hydroxyazobenzene

(2) Orange and p-hydroxyazobenzene

(3) Yellow and p-hydroxyazobenzene

(4) Orange and o-hydroxyazobenzene

72. Choose the incorrect statement

(1) Both La^{3+} and Lu^{3+} ions are colourless

(2) Actinoid contraction is greater from element to element than lanthanoid contraction

(3) Ce(IV) is a good analytical reagent

(4) Both Ce^{4+} and Yb^{2+} are paramagnetic

73. The correct electronic configuration of Gadolinium and Lutetium respectively are

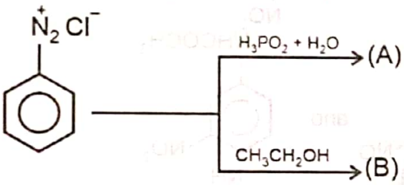
(1) $[Xe]4f^7\,5d^1\,6s^2$ and $[Xe]\,4f^{14}\,5d^1\,6s^2$

(2) $[Xe]4f^8\,6s^2$ and $[Xe]\,4f^{14}\,6s^2$

(3) $[Xe]4f^9\,6s^1$ and $[Xe]\,4f^{13}\,5d^2\,6s^2$

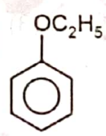
(4) $[Xe]4f^{13}\,5d^1\,6s^2$ and $[Xe]\,4f^7\,5d^1\,6s^2$

74. In the following reaction sequences



Product (A) and (B) respectively are

(1)  and 

(2)  and 

(3)  and 

(4)  and 

75. $KMnO_4$ converts iodide ion in acidic medium and faintly alkaline medium into

(1) Iodate and iodate respectively

(2) Iodine and iodate respectively

(3) Iodine and iodine respectively

(4) Iodate and iodine respectively

76. Given below are two statements.

Statement I : Ca^{2+} and Mg^{2+} ions in hard water form stable complexes with EDTA with different stability constants.

Statement II : Gold can be separated in metallic form from aqueous solution of $[Au(CN)_2]^-$ by the addition of zinc

In the light of above statements choose the correct answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

77. Match List-I with List-II

	List-I Process		List-II Catalyst
a.	Wacker's process	(i)	$TiCl_4 + Al(CH_3)_3$
b.	Manufacture of polythene	(ii)	Fe
c.	Haber process	(iii)	V_2O_5
d.	Oxidation of SO_2 in the manufacture of H_2SO_4	(iv)	$PdCl_2$

Choose the correct option

(1) a(i), b(iv), c(ii), d(iii)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(i), b(ii), c(iii), d(iv)

Space for Rough Work

78. Match List-I with List-II

	List-I Compounds		List-II Isomers
a.	$[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$	(i)	Linkage
b.	$[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$	(ii)	Ionisation
c.	$[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$	(iii)	Solvate
d.	$[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$	(iv)	Coordination

Choose the correct option

(1) a(iii), b(ii), c(i), d(iv)
(2) a(iii), b(i), c(ii), d(iv)
(3) a(iv), b(i), c(ii), d(iii)
(4) a(iv), b(ii), c(i), d(iii)

79. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A) : N–Ethylbenzenesulphonamide is soluble in alkali.

Reason (R) : The hydrogen attached to nitrogen in N-Ethylbenzenesulphonamide is strongly acidic due to the presence of strong electron withdrawing sulphonyl group.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

80. The number of mole of $K_2Cr_2O_7$ required to oxidise 2 moles of Fe^{2+} in acidic medium is

(1) $\frac{3}{4}$
(2) $\frac{1}{6}$
(3) $\frac{1}{3}$
(4) $\frac{2}{5}$

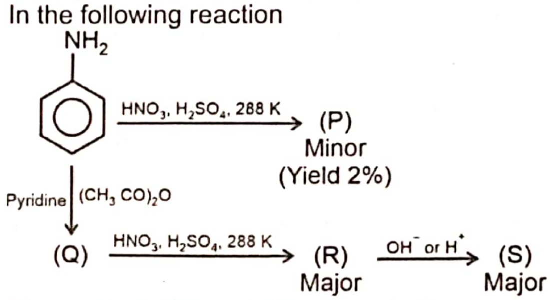
81. Consider the following statements

(a) Haemoglobin is a coordination compound of cobalt.
(b) Chlorophyll is a coordination compound of magnesium.
(c) Vitamin B_{12} is a coordination compound of iron.
(d) Wilkinson catalyst is a coordination compound of rhodium.

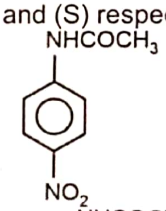
The correct statements are

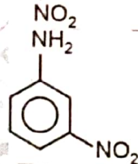
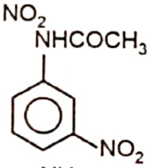
(1) (a) and (c) only
(2) (a) and (b) only
(3) (b) and (d) only
(4) (c) and (d) only

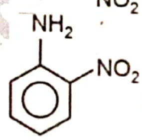
82. In the following reaction

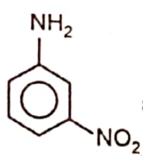
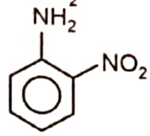


The products (P) and (S) respectively are

(1)  and 

(2)  and 

(3)  and 

(4)  and 

Space for Rough Work

83. Given below are two statements.

Statement I: Aniline reacts with bromine water at room temperature to give a white precipitate of 2, 4, 6-tribromoaniline

Statement II: Aniline does not undergo Friedel-Crafts reaction.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

84. In the following reaction

$[Ni(H_2O)_6]^{2+}_{(aq)}$ + ethane-1, 2-diamine$_{(aq)}$ $\rightarrow$ (A)
(1 equivalent)

$[Ni(H_2O)_6]^{2+}_{(aq)}$ + ethane-1, 2-diamine$_{(aq)}$ $\rightarrow$ (B)
(3 equivalent)

The colour of (A) and (B) respectively are

(1) Pale blue and green (2) Green and purple
(3) Purple and green (4) Pale blue and violet

85. Out of Th, Np, Cf and No, the actinoids which show only +3 oxidation state are

(1) Th and Np (2) Cf and No
(3) Th and Cf (4) Th, Cf and No

86. Given below are two statements.

Statement I : Sodium dichromate is more soluble than potassium dichromate

Statement II : The colour of sodium dichromate and potassium dichromate are orange and yellow respectively.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

87. Consider the following statements about Werner's theory of coordination compounds.

(a) The primary valences are normally ionisable
(b) The secondary valence is equal to the coordination number
(c) Secondary valency of the complex $[Pd(NH_3)_4]Cl_2$ is four

The correct statements are

(1) (a) and (b) only (2) (a), (b) and (c)
(3) (b) and (c) only (4) (a) and (c) only

88. Consider the following species

(a) $[Co(NH_3)_4Cl_2]^+$ (b) $[Co(en)_3]^{3+}$
(c) $[Co(en)_2Cl_2]$ (d) $[Co(NH_3)_3(NO_2)_3]$

Species, those show geometrical isomerism are

(1) (c) and (d) only (2) (a), (c) and (d) only
(3) (a) and (b) only (4) (a), (b) and (c) only

89. Which of the following pairs of ions will have same spin only magnetic moment values within the pair?

(a) Cr^{2+} and Fe^{2+} (b) Zn^{2+} and V^{2+}
(c) Co^{2+} and V^{2+} (d) Cu^{2+} and Ni^{2+}

Choose the correct answer from the options given below

(1) (a) and (c) only (2) (b) and (d) only
(3) (a) and (b) only (4) (a), (b) and (c) only

90. Consider the following statements

(a) Cr_2O_3 is an amphoteric oxide
(b) Mn_2O_7 is a covalent green oil
(c) Manganate ion is diamagnetic in nature

The correct statements are

(1) (a) and (c) only (2) (b) and (c) only
(3) (a) and (b) only (4) (a), (b) and (c)

Space for Rough Work

[BOTANY]

91. Read the following statements and select the **correct** option

Statement A: The three-dimensional structure of tRNA was proposed to be an inverted L.

Statement B: Severo Ochoa enzyme is helpful in polymerising RNA with defined sequence in a template independent manner.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

92. RNA polymerase III transcribes all of the following RNAs, **except**

(1) tRNA

(2) 5.8 S rRNA

(3) SnRNA

(4) 5S rRNA

93. The charging of tRNA refers to the

(1) Linking of amino acid to their cognate tRNA using ATP

(2) Binding of tRNA with smaller subunit of ribosomes in eukaryotes

(3) Movement of tRNA on the ribosome

(4) Binding of tRNA with rRNA using GTP

94. Match the Column I with Column II and select the **correct** option.

	Column I		Column II
a.	Frederick Griffith	(i)	Preparation of polynucleotide using synthetic DNA
b.	Avery, MacLeod and McCarty	(ii)	Transforming principle
c.	Har Gobind Khorana	(iii)	Semiconservative replication of DNA
d.	Meselson and Stahl	(iv)	Identifying biochemical nature of transforming principle

(1) a(iv), b(ii), c(i), d(iii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(i), c(iv), d(ii)

95. Select the **incorrectly** matched pair

(1)	Euchromatin	–	Transcriptionally active
(2)	Heterochromatin	–	Loosely packed chromatin
(3)	Nucleosome	–	Gives a 'beads on string' like appearance
(4)	Nuclein	–	Acidic substance in the nucleus

96. Uracil is a

(1) Nucleotide

(2) Nucleoside

(3) Methylated form of thymine

(4) Nitrogenous base found in RNA

Space for Rough Work

97. Read the following four statements (A - D)

(A) A promoter is said to be located towards 3′-end (upstream) of the structural gene w.r.t the polarity of coding strand.

(B) The terminator is located towards 3′-end (downstream) of the coding stand.

(C) There are additional regulatory sequences that may be present further upstream or downstream to the promoter.

(D) The UTRs are present only before start codon at 5′-end in eukaryotic mRNA.

In the light of above statements select the **correct** option.

(1) Only (A) and (C) are correct
(2) Only (B) and (C) are correct
(3) (A), (B) and (D) are correct
(4) (A), (C) and (D) are correct

98. Select the **incorrectly** matched pair.

(1) BAC – Bioreactive artificial chromosomes
(2) YAC – Yeast artificial chromosomes
(3) SNPs – Single nucleotide polymorphism
(4) VNTR – Variable number of tandem repeats.

99. Select the **incorrect** statement w.r.t salient features of human genome.

(1) Almost all (99.9%) nucleotide bases are exactly the same in all people.
(2) Scientists have identified about 1.4 million locations where single base DNA differences occur in humans.
(3) Repetitive sequences are thought to have direct coding functions.
(4) Chromosome 1 has most gene (2968) and the Y has the fewest (231)

100. Read the following statements and select the **correct** option.

Statement A: DNA is preferred over RNA for storage of genetic information.

Statement B: RNA can directly code for the synthesis of proteins, and it is better material for the transmission of genetic information.

(1) Both the statements A and B are correct
(2) Both the statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

101. A molecule that can act as genetic material must fulfill all the following criteria, **except**

(1) It should be able to generate its replica
(2) It should be able to express itself in the form of 'Mendelian characters'
(3) It should not provide the scope for slow changes that may lead to evolution
(4) It should be stable chemically and structurally

102. Following is the sequence of DNA template strand.
3′ TACGTAGCTAGGATC 5′.
What will be the sequence of tRNAs anticodons that translate the mRNA transcribed from this segment?

	1st tRNA	2nd tRNA	3rd tRNA	4th tRNA
(1)	3′UAC5′	3′GUA5′	3′GCU5′	3′AGG5′
(2)	5′UAC3′	5′GUA3′	5′GCU3′	5′AGG3′
(3)	3′AUG5′	3′CAU5′	3′CGA5′	3′UCC5′
(4)	5′AUG3′	5′CAU3′	5′CGA3′	5′UCC3′

103. Genetic code is said to be degenerate because

(1) One codon codes for only one amino acid
(2) Codon is read on mRNA in a contiguous fashion
(3) Three codons do not code for any amino acid
(4) Some amino acids are coded by more than one codon.

Space for Rough Work

104. Two nucleotides are linked through ______ linkage to form a dinucleotide.
Fill in the above blank by selecting the **correct** option.
(1) 1′ - 9′ glycosidic
(2) 3′ - 5′ phosphodiester
(3) N-glycosidic
(4) 5′ - 3′ phosphoester

105. The genome of bacteriophage lambda has
(1) 48502 nucleotides (2) 3.3×10^9 base pairs
(3) 5386 nucleotides (4) 48502 base pairs

106. The experimental proof that DNA replicates in a semiconservative manner was first shown using
(1) *Escherichia coli* (2) Bacteriophage λ
(3) A unicellular fungi (4) A blue green alga

107. Transcription unit has all the given components, **except**
(1) Promoter (2) Inducer
(3) Structural gene (4) Terminator

108. Which of the following ribosomal RNA catalyzes the formation of peptide bonds in bacteria?
(1) 5S rRNA (2) 18S rRNA
(3) 23S rRNA (4) 5.8S rRNA

109. During unwinding of DNA, there is formation of supercoils, opposite to the replication fork. In nuclei of eukaryotes, the tension so developed in the DNA is released by the action of the enzyme
(1) DNA gyrase (2) Helicase
(3) DNA polymerase (4) Topoisomerase

110. An *E. coli* cell is transferred from N^{15} to N^{14} medium and is allowed to multiply for 4 generations. What will be the fraction of the hybrid DNA extracted from the *E. coli* culture?
(1) 25% (2) 50%
(3) 75% (4) 12.5%

111. A mutation in *E. coli* changes the *lac* operon at the site indicated in the below given figure that causes the inactivity of the gene product.

p	*i*	*P*	*o*	*z* *y* *a*

Mutation
point

The consequence of this mutation will be that
(1) The cell would continuously produce regulatory gene product
(2) The structural genes of the *lac* operon will not be expressed in the presence of allolactose
(3) The cell would continuously produce structural gene products even in the absence of inducer
(4) The RNA polymerase will not be able to transcribe the gene due to inactivation of operator

112. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A) : Chemically DNA is less reactive and structurally more stable when compared to RNA.
Reason (R) : DNA have thymine in place of uracil along with an additional –OH group present at 2′ position in the sugar.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(2) Both (A) and (R) are true and (R) is the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

113. Bacterial species are used in
(1) Production of alcohol at commercial scale
(2) Preparation of dough for dosa and idli
(3) Preparation of bread
(4) Ripening of Camembert cheese

Space for Rough Work

114. The biocontrol agents used for controlling aphids and butterfly caterpillars respectively are
(1) Dragonflies and Ladybird beetle
(2) *Trichoderma* and Ladybird beetle
(3) Ladybird beetle and *Bacillus thuringiensis*
(4) Baculoviruses and Dragonflies

115. In order to save the major rivers of our country, Ganga action plan and Yamuna action plan was initiated by.
(1) The Ministry of agriculture and farmer's welfare
(2) Khadi and Village Industries Commission
(3) Indian Agricultural Research Institute
(4) The Ministry of Environment and Forests

116. Select the **odd** one out w.r.t. primary treatment of waste water.
(1) It is a physico-chemical process
(2) Sequential filtration is done
(3) Sedimentation of grit occurs
(4) Vigorous growth of microbes into flocs does not occur in this step

117. Which of the following is used as a clot buster for patients who have undergone myocardial infarction?
(1) Cellulase (2) Protease
(3) HMG CoA reductase (4) Streptokinase

118. Match the Column I with Column II and select the **correct** option.

	Column I		Column II
a.	Swiss cheese	(i)	*Caryota urens*
b.	Cyclosporin A	(ii)	*Monascus purpureus*
c.	Toddy	(iii)	*Propionibacterium shermanii*
d.	Statins	(iv)	*Trichoderma polysporum*

(1) a(iii), b(i), c(ii), d(iv)
(2) a(iv), b(ii), c(iii), d(i)
(3) a(iii), b(iv), c(i), d(ii)
(4) a(ii), b(i), c(iv), d(iii)

119. State True **(T)** or False **(F)** to the given statements:
(a) In India, *Aulosira* is most active symbiotic nitrogen fixer in rice field.
(b) Free-living *Trichoderma* species are very common in the shoot ecosystems.
(c) Penicillin was extensively used to treat American soldiers wounded in World War II.
(d) Baculoviruses have negative impacts on birds, fishes and non-target insects.

Choose the **correct** option :

	(a)	(b)	(c)	(d)
(1)	T	F	F	T
(2)	T	T	F	T
(3)	T	T	T	F
(4)	F	F	T	F

120. Butyric acid used for making rancid butter is obtained from
(1) *Acetobacter* (2) *Clostridium*
(3) *Aspergillus* (4) *Saccharomyces*

121. Identify the **incorrect** statement w.r.t. biogas plant.
(1) A floating cover is placed over the slurry, which keeps on sinking as the gas is produced in the tank due to microbial activity
(2) It has an outlet, which is connected to a pipe to supply biogas to nearby houses
(3) The spent slurry is removed through another outlet and may be used as fertiliser
(4) The technology of biogas production was developed in India mainly by the efforts of IARI and KVIC

Space for Rough Work

122. Biochemical oxygen demand

(a) Refers to the amount of the oxygen that would be released if all the organic matter in one liter of water were oxidised by bacteria.

(b) Is a measure of the organic matter present in the water.

(c) Is lesser in the waste water where polluting potential is low.

The **correct** one(s) is/are

(1) Only (a)
(2) (b) and (c)
(3) Only (c)
(4) (a) and (c)

123. Cyanobacteria

(1) Form mycorrhizal association with higher plants
(2) Are used as biocontrol agents
(3) Serve as an important biofertilizer in rice fields
(4) Can fix atmospheric nitrogen but they are unable to fix atmospheric CO_2

124. Select the **incorrect** statement from the following.

(1) Integrated pest management (IPM) programme is desirable when beneficial insects are being conserved
(2) IPM involves acceptable combination of only physical and chemical methods to limit the harmful effects of crop pests
(3) When an ecologically sensitive area is needs to be treated then integrated pest management is most desirable option
(4) The overall objective of IPM is to create and to maintain situations in which insects are prevented from causing significant damage to crops

125. To be a part of an integrated pest management, a biocontrol agent

(1) Should always be symbiotic as well as free living N_2-fixing organism
(2) Cannot be of narrow spectrum application
(3) Should be species-specific and have no negative impacts on non-target organism
(4) Cannot be pathogenic to any of the organisms

126. During bread making, the puffed-up appearance of dough is due to production of

(1) Carbon dioxide
(2) Methane
(3) Hydrogen sulphide
(4) Nitrogen

127. The bacteria commonly found in anaerobic sludge during sewage treatment and also play an important role in nutrition of cattle is

(1) *Azospirillum*
(2) *Methanobacterium*
(3) *Azotobacter*
(4) *Rhizobium*

128. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

Assertion (A) : LAB play very beneficial role in checking disease causing microbes in our stomach.

Reason (R) : LAB improves the nutritional quality of curd by increasing vitamin B_{12}.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

129. RNA splicing

(1) Is the process of removal of exons from the mRNA
(2) Involves endonucleolytic cleavage at splice junctions of hnRNA by SnRNPs
(3) Occurs at 3′ end in template independent manner
(4) Is a primary step of post-transcriptional processing in *E. coli*

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130. Which of the following enzymes has the capability to synthesise DNA over RNA in a template dependent manner?

(1) DNA dependent RNA polymerase

(2) RNA dependent RNA polymerase

(3) Reverse transcriptase

(4) Transacetylase

131. According to Watson and Crick, the DNA helix

(1) Backbone is constituted by sugar-phosphate and the bases project outside

(2) Consists of two anti-parallel polynucleotide chains where 3′-end has a free phosphate moiety

(3) Is twisted helically in right-handed fashion and each turn has roughly 10 bp

(4) Does not get stabilised by the H-bond between purines and pyrimidines.

132. Under which of the following conditions will there be frameshift mutation of the following mRNA?

5′ AACAUGCGGUGAUAUU 3′

(1) Insertion of poly A tail at 3′ end

(2) Deletion of UAU from 13^{th}, 14^{th} and 15^{th} positions respectively

(3) Insertion of A just after 1^{st} G from 5′-end

(4) Deletion of A, U and G from 4^{th}, 5^{th} and 6^{th} positions respectively

133. Read the following statements and mark them as true (T) or false (F).

A. Structural genes in eukaryotes are mostly monocistronic.

B. The RNA polymerase (core enzyme) is not capable of catalysing the process of elongation of RNA.

C. Both transcription and translation take place in the cytosol of the eukaryotic cell.

D. All tRNA molecules have a guanine residue at its 5′ terminal end

Choose the **correct** option.

	A	B	C	D
(1)	T	F	F	F
(2)	F	T	F	T
(3)	T	F	F	T
(4)	F	F	T	F

134. Polymorphism in DNA sequence is the basis of

(a) Genetic mapping of human genome.

(b) DNA fingerprinting

(c) DNA polymerisation in all organisms

The **correct** one(s) is/are

(1) Only (a)

(2) Only (a) and (b)

(3) Only (c)

(4) Only (a) and (c)

135. Which of the following statements does **not** represent the accurate reason for not copying both the strands of DNA during transcription?

(1) One segment of the DNA would be coding for two different proteins

(2) The two RNA molecules if produced simultaneously would be complimentary to each other

(3) Formation of dsRNA will prevent translation

(4) Sequence of amino acids coded by both RNA molecules will be same

Space for Rough Work

[ZOOLOGY]

136. All of the following plants produce certain substances which cause hallucinations at certain doses, **except**
(1) *Erythroxylum coca* (2) *Atropa belladona*
(3) *Datura* (4) *Papaver somniferum*

137. The immune response that is primarily responsible for the graft rejection, is mediated by cells that
(1) Belong to cytokine barrier of innate immunity
(2) Secrete molecules represented as H_2L_2
(3) Themselves do not secrete antibodies
(4) Mainly mediate humoral immune response

138. If a person is infected with the deadly microbe that causes tetanus, passive immunisation is preferred over active, because
(1) Former allows production of antibodies in the body
(2) In former, directly attenuated pathogen is administered
(3) Latter is more effective and provides quick immune response
(4) Latter is slow and takes time to give its full effective response

139. All of the following are ancestors of mammals, **except**
(1) Synapsids (2) Therapsids
(3) Thecodonts (4) Pelycosaurs

140. Select the **incorrect** statement w.r.t antibody.
(1) Antigen binding site is present on N-terminal amino acids of heavy and light chains.
(2) Antibodies are found in the blood.
(3) Each monomeric antibody molecule contains two types of peptide chains.
(4) Overall, there are 12 disulfide bonds present in a monomeric antibody molecule.

141. The chemicals released to cause allergy are
(1) Steroids and heparin from basophils
(2) Histamine and serotonin from mast cells
(3) Heparin and serotonin from thrombocytes
(4) Histamine and adrenaline from basophils

142. S.L. Miller observed the formation of ________ in the spark discharge experiment performed by him.
Select the **correct** option to fill in the blank.
(1) Amino acids
(2) Sugars
(3) Nitrogenous bases
(4) Pigments

143. How many of the organs mentioned in the box below are involved in differentiation of immature lymphocytes into antigen-sensitive lymphocytes in humans?

Bone marrow, Spleen, Tonsils, Thymus, Lymph nodes

Choose the **correct** option.
(1) Three (2) Four
(3) Two (4) Five

144. A large bean-shaped, secondary lymphoid organ which acts as a filter of the blood by trapping blood-borne microorganisms, mainly contains
(1) Monocytes and Interferons
(2) Thrombocytes and NK cells
(3) Lymphocytes and Phagocytes
(4) Megakaryocytes and Lymphocytes

145. Transmission of HIV infection can occur by all of the following ways, **except**
(1) Sexual contact with the infected person
(2) Transfusion of contaminated blood
(3) Using infected needles in case of intravenous drug injection
(4) By hugging or shaking hands with an infected person

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146. Interferons secreted from virus-infected cells belong to ________ barrier of innate immunity.

Select the **correct** option to fill in the blank.

(1) Physical
(2) Physiological
(3) Cellular
(4) Cytokine

147. The tendency of the human body to manifest a withdrawal syndrome, if regular dose of drug/alcohol is abruptly discontinued, is

(1) Tolerance
(2) Dependence
(3) Addiction
(4) Euphoria

148. Read the following statements.

Statement (A) : Heroin, commonly called smack is obtained by acetylation of morphine.

Statement (B) : Cocaine, commonly called coke or crack has a potent stimulating action on CNS.

Select the **correct** option.

(1) Both statements (A) and (B) are incorrect
(2) Only statement (A) is correct
(3) Both statements (A) and (B) are correct
(4) Only statement (B) is correct

149. Select the **correct** option w.r.t AIDS.

(1) Macrophages act like HIV-factory.
(2) Macrophages cannot survive while viruses are being replicated and released.
(3) Widal test is the widely used diagnostic test.
(4) Treatment with anti-retroviral drugs are fully effective and can cure the disease.

150. **Assertion (A):** AIDS patient suffers from infections that could have been otherwise overcome such as those due to *Mycobacterium* and *Toxoplasma*.

Reason (R): There is decrease in the number of helper T-lymphocytes in the body of an AIDS patient below a certain level.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are false
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) (A) is true but (R) is false

151. Select the **correct** option to complete the analogy.

Natural selection : Charles Darwin : : ________ : Lamarck

(1) Branching descent
(2) Saltation
(3) Use and disuse of organs
(4) Founder effect

152. Select the **incorrect** match among the following.

(1)	Physical barrier of innate immunity	–	Mucus coating of the epithelium lining the respiratory tract
(2)	Primary immune response	–	Low intensity immune response as compared to anamnestic response
(3)	Cellular barrier of innate immunity	–	Macrophages in blood and B-lymphocytes in tissues
(4)	Secondary immune response	–	Anamnestic immune response

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153. Choose the **incorrect** point of difference between benign and malignant tumors.

	Benign tumor	Malignant tumor
(1)	Slow growth	Rapid growth
(2)	Does not exhibit metastasis	Exhibits metastasis
(3)	Does not show contact inhibition	Shows contact inhibition
(4)	Does not invade nearby tissue	Invade the nearby tissue

154. Which of the following theories attempts to explain to us the origin of universe?
(1) Panspermia theory
(2) The theory of spontaneous generation
(3) The Big Bang theory
(4) Chemical evolution theory

155. In 1953, S.L. Miller, an American scientist created electric discharge in a closed flask containing
(1) O_2, NH_3, CH_4 and water vapour at 800°C
(2) CH_4, H_2, NH_3 and water vapour at 800°C
(3) NH_3, O_2, CH_4 and water vapour at 500°C
(4) CH_4, H_2, NH_3 and water vapour at 600°C

156. All of the following indicates common ancestry, **except**
(1) Wings of butterfly and that of birds
(2) Forelimbs of cheetah and bat
(3) Vertebrate brains
(4) The thorn of *Bougainvillea* and tendrils of *Cucurbita*

157. About 50% of the lymphoid tissue in human body is/are contributed alone by
(1) Lymph nodes
(2) Peyer's patches
(3) MALT
(4) Tonsils

158. Select the **correct** statement.
(1) Darwinian variations are random and directionless.
(2) Hugo de Vries believed that only minor heritable variations cause evolution.
(3) Hugo de Vries brought forth the idea that large difference arise suddenly in a population.
(4) Darwin believed that sudden mutation could cause evolution.

159. Select the option which represents **correct** sequence of evolution of plant forms through geological periods.
(1) Psilophyton → Seed ferns → Gnetales → Progymnosperms
(2) Chlorophyte ancestors → Tracheophyte ancestors → Ferns → Conifers
(3) Progymnosperms → Seed ferns → Dicots → Monocots
(4) Sphenopsids → Herbaceous lycopods → Zosterophyllum → Psilophyton

160. When migration of a section of population happens multiple times to another place and population, then this phenomenon would be called
(1) Gene migration (2) Gene flow
(3) Genetic drift (4) Genetic equilibrium

161. The lobed organ, located near the heart and beneath the breastbone, which is quite large at the time of birth but keeps reducing in size with age, is
(1) Bone marrow (2) Thymus
(3) Spleen (4) Appendix

162. Read the following carefully.
(a) They probably lived in East African grasslands, about two mya.
(b) They hunted with stone weapons.
(c) They essentially ate fruits.
All of the above characteristics hold **true for**
(1) *Dryopithecus* (2) *Australopithecines*
(3) *Homo sapiens* (4) *Homo erectus*

Space for Rough Work

163. Which among the following land reptiles was about 20 feet in height and had huge fearsome dagger-like teeth?

(1) *Triceratops* (2) *Tyrannosaurus rex*
(3) *Stegosaurus* (4) *Brachiosaurus*

164. **Assertion (A):** The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull.

Reason (R): Chimpanzees are non-primates whereas humans are primates.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

165. Complete the analogy by selecting the **correct** option.

Coelacanth : Lobefin : : *Ichthyosaurs* : ________

(1) Mammal (2) Fish-like reptile
(3) First amphibian (4) Dinosaur

166. All of the following exhibit divergent evolution, **except**

(1) Marsupial mole and Banded anteater
(2) Koala and wombat
(3) Lemur and Spotted cuscus
(4) Flying phalanger and Tasmanian wolf

167. If the sum total of all allelic frequencies in a population deviates from 1, this means that

(1) Allelic frequencies are constant from one generation to another
(2) Gene pool of this population is constant
(3) Evolution is not occurring in this population
(4) Population is not in genetic equilibrium

168. Read the following statements.

Statement (A): Evolution is not a directed process in the sense of determinism.

Statement (B): Evolution is a stochastic process based on chance events in nature and the chance mutation in the organisms.

Select the **correct** option.

(1) Both statements (A) and (B) are correct
(2) Only statement (A) is correct
(3) Both statements (A) and (B) are incorrect
(4) Only statement (B) is correct

169. Which of the following is true regarding industrial melanism?

(1) Before industrialisation, there were more melanised moths as compared to white- winged moths.
(2) Presence of lichens indirectly promoted the growth of population of melanised moths.
(3) After industrialisation, there were more white-winged moths as compared to melanised moths.
(4) No variant of moth was completely wiped out.

170. All of the following are examples that exist because of natural selection and not by anthropogenic action, **except**

(1) Darwin's finches of Galapagos islands
(2) Elongated necks of giraffes
(3) Dogs of different breeds
(4) Australian marsupials and placental mammals

171. Which among the following scientists contributed in providing the embryological support for evolution?

(1) Hugo deVries
(2) Ernst Heckel
(3) Hardy and Weinberg
(4) Oparin and Haldane

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172. Match column I and column II w.r.t diseases and vectors/carriers of their pathogens.

	Column I		Column II
a.	Malaria	(i)	*Culex*
b.	Dengue	(ii)	House fly
c.	Amoebiasis	(iii)	*Anopheles*
d.	Filariasis	(iv)	*Aedes*

Select the **correct** option.

(1) a(i), b(ii), c(iv), d(iii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(iv), c(ii), d(i) (4) a(i), b(iii), c(iv), d(ii)

173. Comprehend the given statements.

(a) The first organisms that invaded land were plants.
(b) Bryophytes are the direct descendants of Psilophyton.
(c) About 85 mya, the dinosaurs suddenly disappeared from the Earth.
(d) The first mammals were like shrews.

How many of the above given statements is/are **incorrect**?

(1) One (2) Two
(3) Three (4) Four

174. Select the **correct** match w.r.t diseases and the category to which their causative agents belong to.

(1)	Filariasis	Helminth
(2)	Typhoid	Protozoa
(3)	Polio	Arthropod
(4)	Amoebic dysentery	Nematode

175. Select the **incorrect** statement.

(1) The universe is almost 20 billion years old.
(2) According to theory of special creation, the Earth is about 4000 years old.
(3) Darwinian fitness refers ultimately and only to reproductive fitness.
(4) Flippers of penguins and that of dolphins are examples of homologous organs.

176. Drugs like marijuana, hashish, charas and ganja are obtained from

(1) *Erythroxylum coca*
(2) *Atropa belladona*
(3) *Cannabis sativa*
(4) *Papaver somniferum*

177. All of the following are consequences of smoking in human body, **except**

(1) Increased CO content in blood
(2) Reduction in the concentration of haem-bound oxygen
(3) Increased incidence of cancer of lungs
(4) Cannot cause gastric ulcer

178. Which among the following are not harmful to human cells under normal conditions but when activated, could lead to oncogenic transformation of the cells?

(1) Sporozoites/cellular oncogenes
(2) Proto-oncogenes/cellular oncogenes
(3) Oncogenic viruses/Merozoites
(4) Viral oncogenes and trophozoites

179. How many of the following is/are the side effects of the use of anabolic steroids in females?

(a) Masculinisation
(b) Increased aggressiveness
(c) Deepening of voice
(d) Mood swings

Select the **correct** option.

(1) Three (2) Two
(3) Four (4) One

180. Which of the following types of natural selection operates when more individuals acquire peripheral character value at both ends of the distribution curve?

(1) Stabilising (2) Directional
(3) Disruptive (4) Cyclic

❑ ❑ ❑

Space for Rough Work

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